

Data Visualization and Storytelling

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MSc Business Analytics



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SESSION 4: Telling the truth about data, graphical integrity, Power BI practice

Graphical integrity **best practices**

- Show the data
- Remember that the goal of a visual display is to **inform** about some data – not to divert
- Do not hide it or over-process it
- Induce the audience to think about **substance**, not about methodology or graphic production
- Avoid **distorting** the data

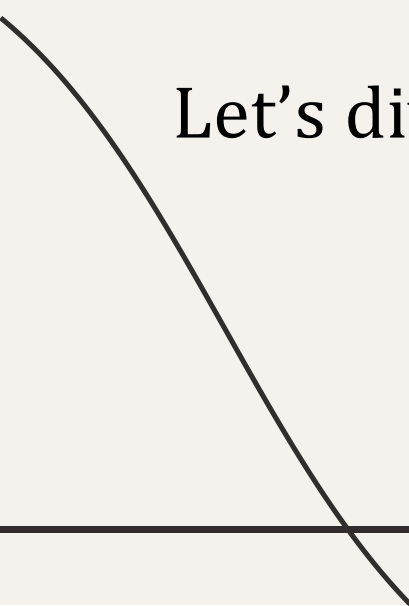


Graphical integrity

1 Telling the truth about the data: The representation of numbers, as physically measured on the surface should be directly proportional to the numerical quantities represented.

This is about **how accurately visual elements represent data.**

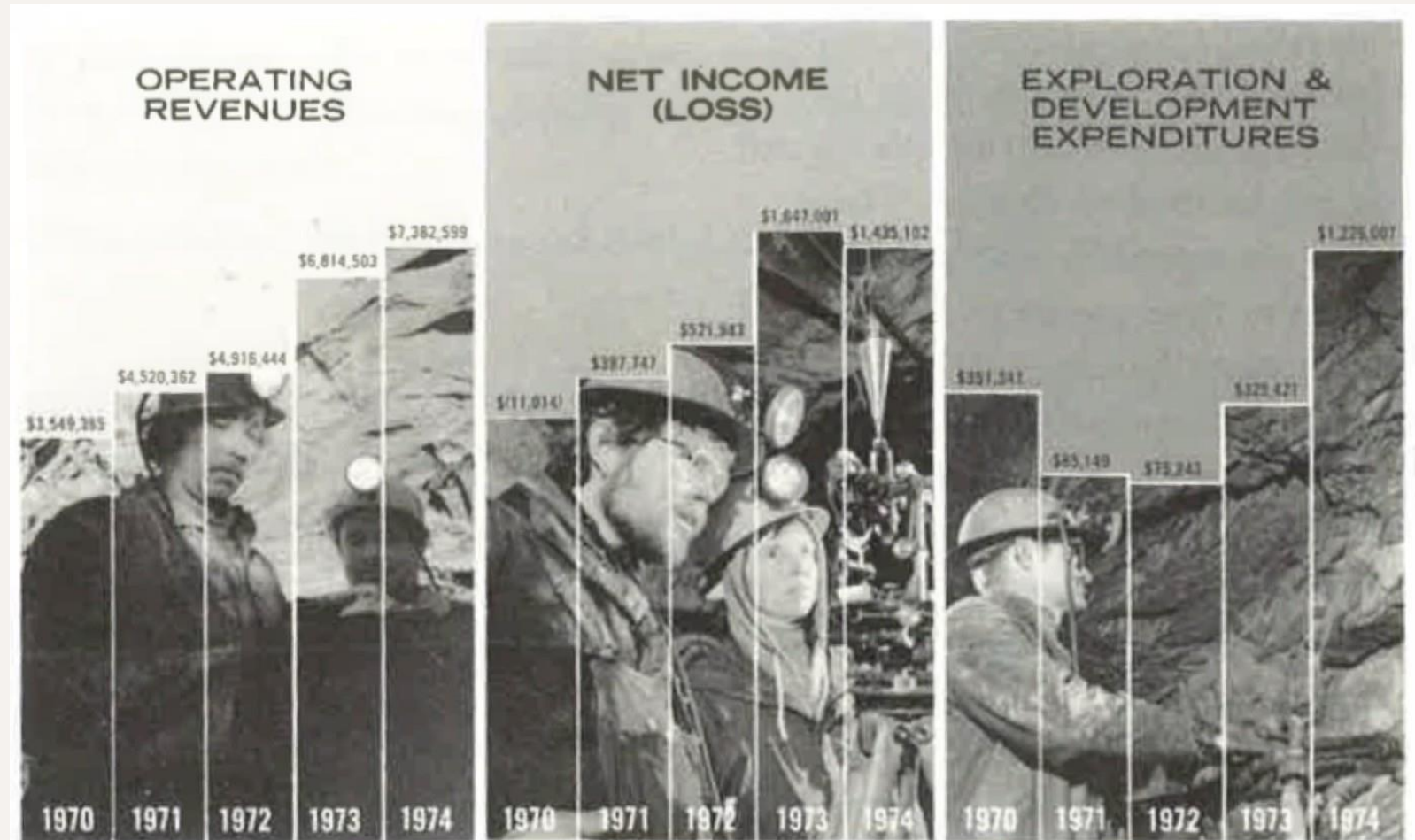
Let's dive into an example.





Graphical integrity

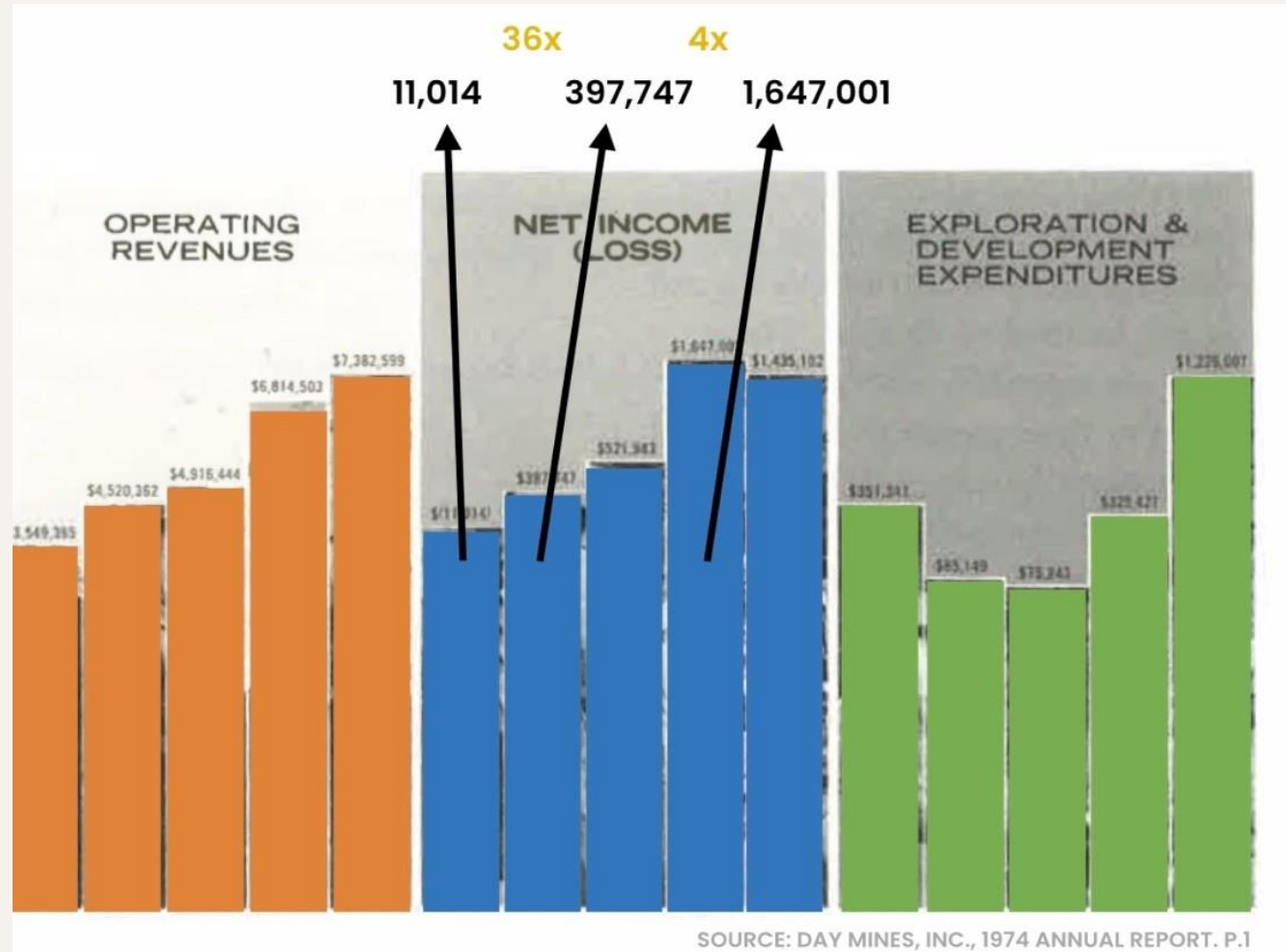
1 Telling the truth about the data



SOURCE: DAY MINES, INC., 1974 ANNUAL REPORT. P.1

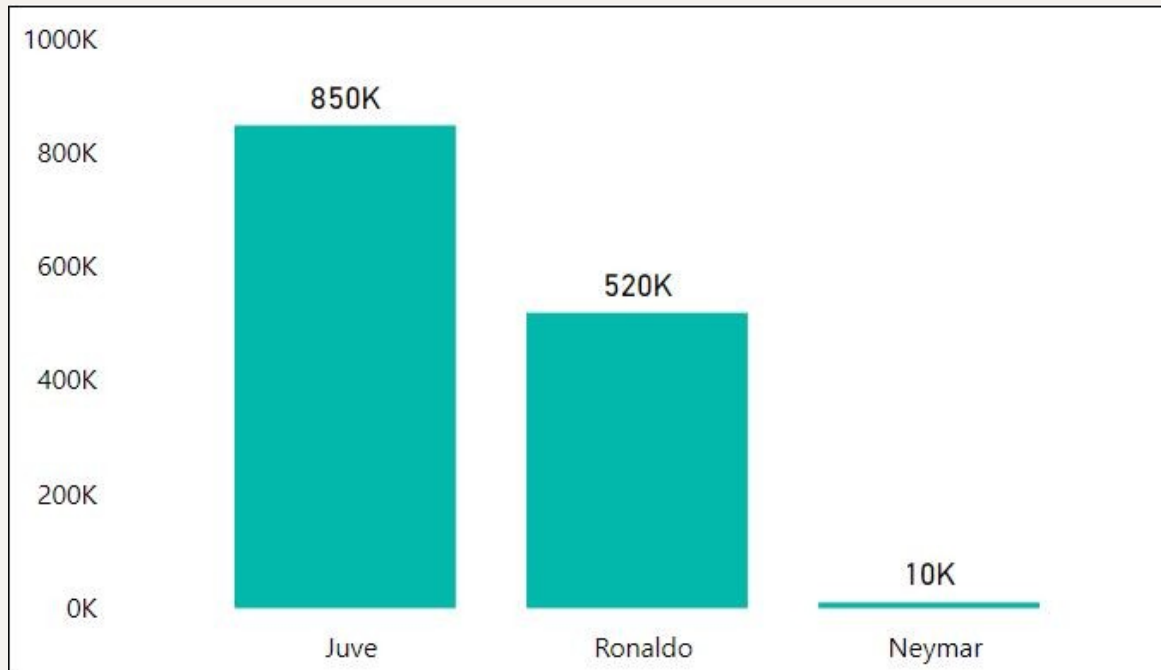
Graphical integrity

1 Telling the truth about the data



Graphical integrity

1 Telling the truth about the data



IFTV

STATS VIA YAHOO AND
FOX SPORTS

MARKETING MACHINE

TOTAL JUVE
JERSEY'S SOLD
IN 2016

850K

RONALDO
JERSEYS SOLD
IN THE FIRST DAY

520K

NEYMAR
JERSEYS SOLD
FROM BARCA
TO PSG

10K



Graphical integrity

1 Telling the truth about the data

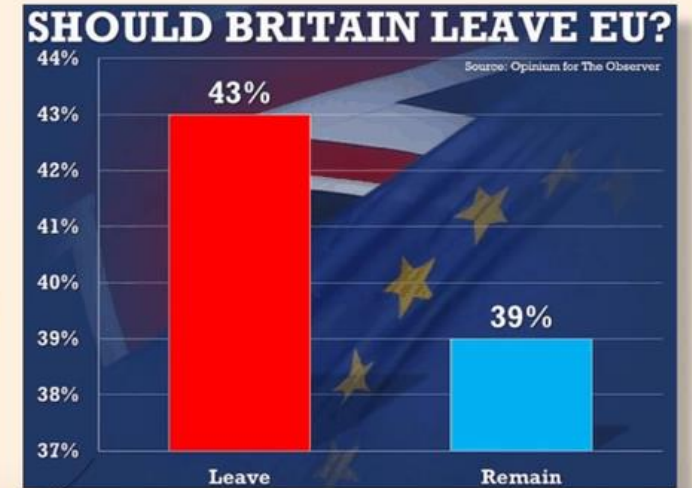


Cherry Picking

Selecting only the data that supports your conclusion — while ignoring data that doesn't — is a form of graphical dishonesty.

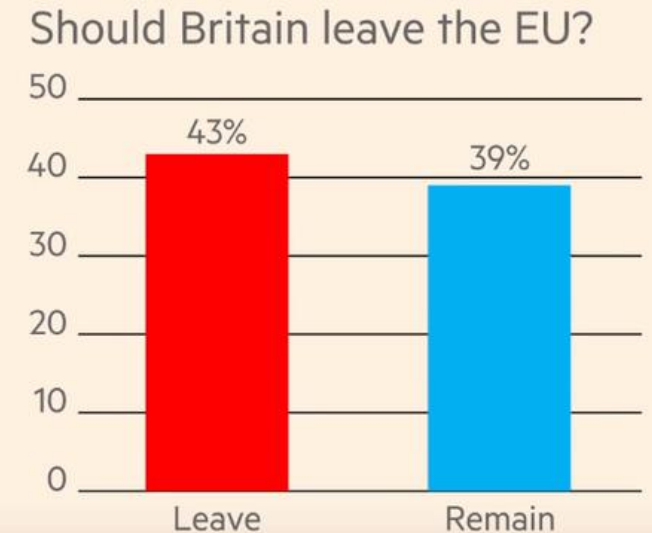
Graphics that are accurate but misleading

Baseline should start at zero, not 37



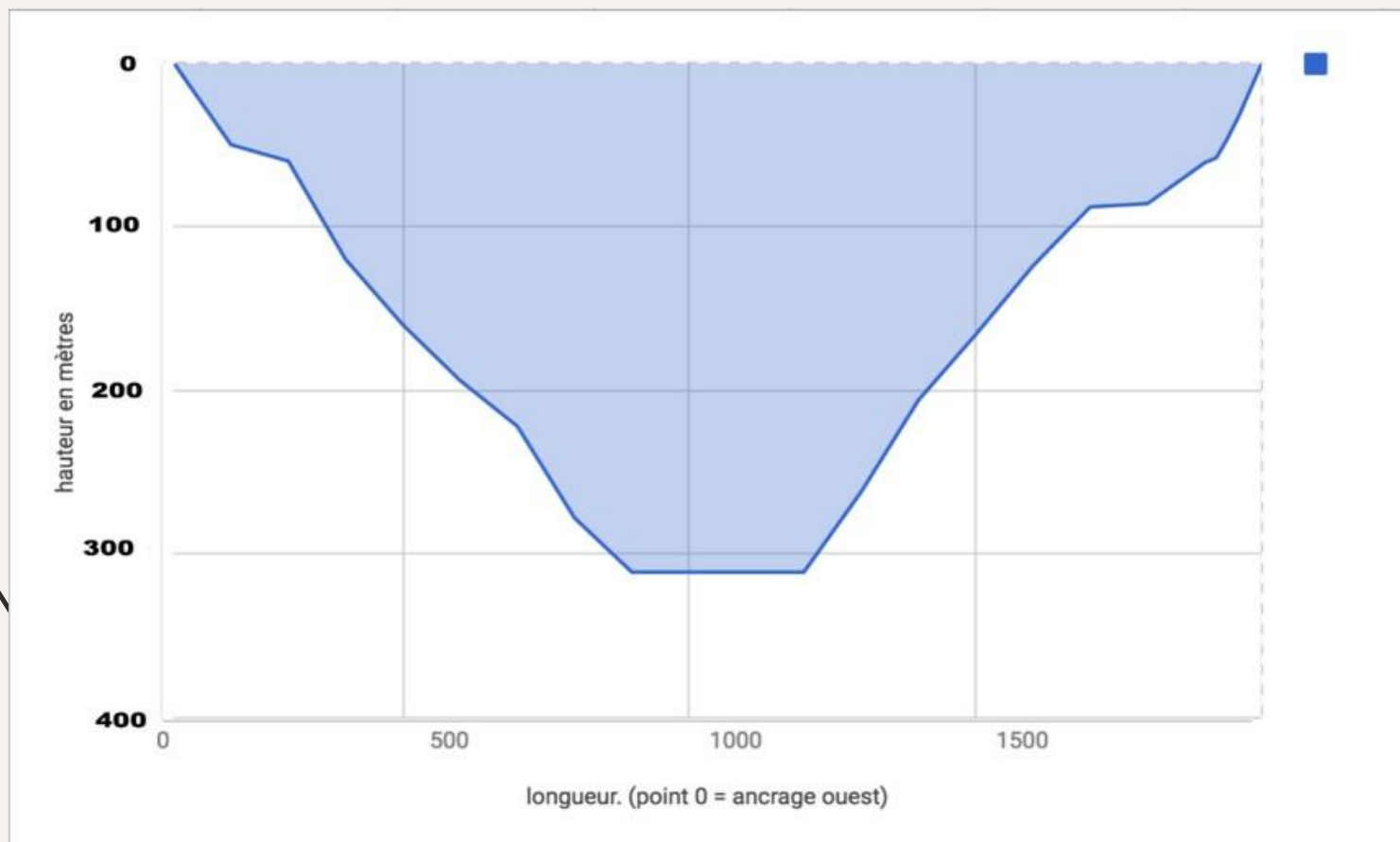
Graphics that are accurate but misleading

A better chart of the same data



Graphical integrity

1 Telling the truth about the data



Inverted Axis

Flipping the y-axis makes an increase look like a decrease. The audience reads downward as 'going down.'

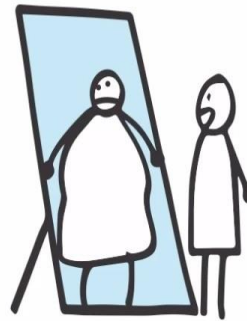
Graphical integrity

1 Telling the truth about the data

Let's Look at the Lie Factor:

Value to describe the relation between the **size of effect shown in a graphic** and the **size of effect shown in the data**

$$\text{Lie Factor} = \frac{\text{size of effect shown in the graphic}}{\text{real size of effect in the data}}$$



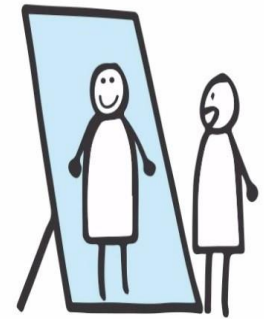
Lie Factor > 1

The representation of the data is over-representing the real effect of the data



Lie Factor < 1

The representation of the data is under-representing the real effect



Lie Factor $= 1$

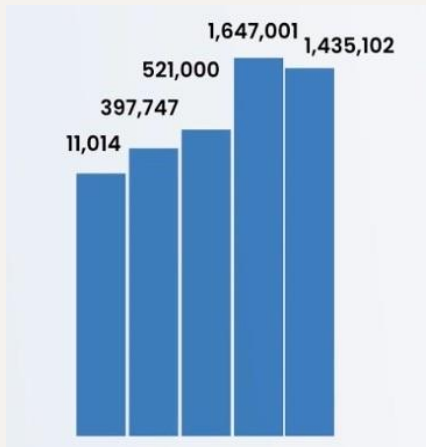
The representation of the data is true to the real quantitative data

Graphical integrity

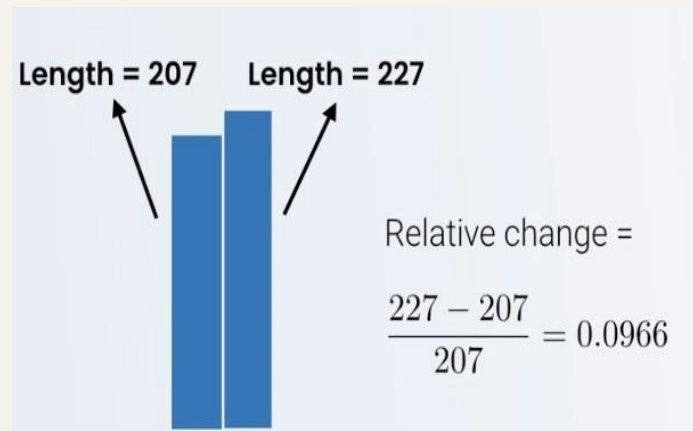
1 Telling the truth about the data

Calculating the Lie Factor

Original Plot



How it appeared in figure



Effect in the data

Data of bar 1 = 11,014

Data of bar 2 = 397,747

$$\text{Relative change} = \frac{397747 - 11014}{11014} = 35.113$$

Graphical integrity

1 Telling the truth about the data

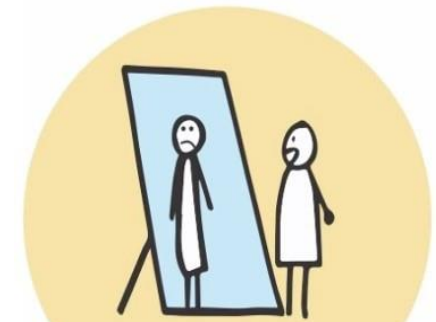
Calculating the Lie Factor

Effect in the data

Data of bar 1 = 11,014

Data of bar 2 = 397,747

$$\text{Relative change} = \frac{397747 - 11014}{11014} = 35.113$$



Lie Factor < 1

The representation of the data is under-representing the real effect

$$\text{Lie Factor} = \frac{\text{graphic effect}}{\text{data effect}} = \frac{0.0966}{35.113} = 0.00275$$



Graphical integrity

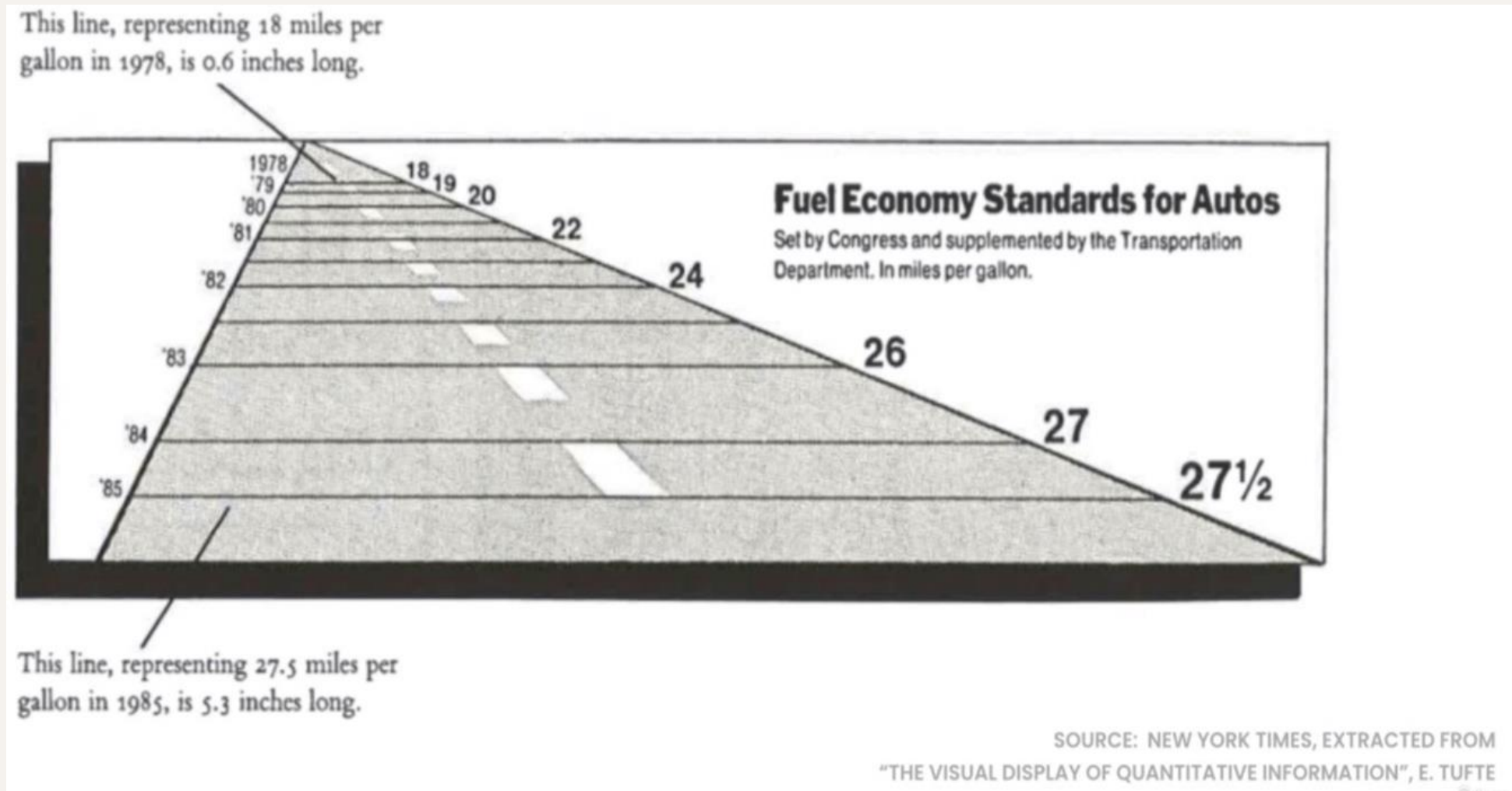
1 Telling the truth about the data

Calculating the Lie Factor: Your turn

A newspaper reported that the US Congress and the Department of Transportation had set a series of fuel economy standards to be met by automobile manufacturers, beginning with 18 miles per gallon in 1978 and moving in steps up to 27.5 by 1985, an increase of 53%.



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Graphical integrity

1 Telling the truth about the data

Calculating the Lie Factor: Your turn

A newspaper reported that the US Congress and the Department of Transportation had set a series of fuel economy standards to be met by automobile manufacturers, beginning with 18 miles per gallon in 1978 and moving in steps up to 27.5 by 1985, an increase of 53%.

The magnitude of the change from 1978 to 1985 is shown in the graph by the relative lengths of the two lines:
$$5.3 - 0.6 / 0.6 \times 100 = 783\%$$

Thus the numerical change of 53% is presented by some lines that changed 783 percent, yielding:

$$\text{Lie factor} = 783 / 53 = 14.8$$



Lie Factor > 1

The representation of the data is over-representing the real effect of the data



Graphical integrity

1 Telling the truth about the data

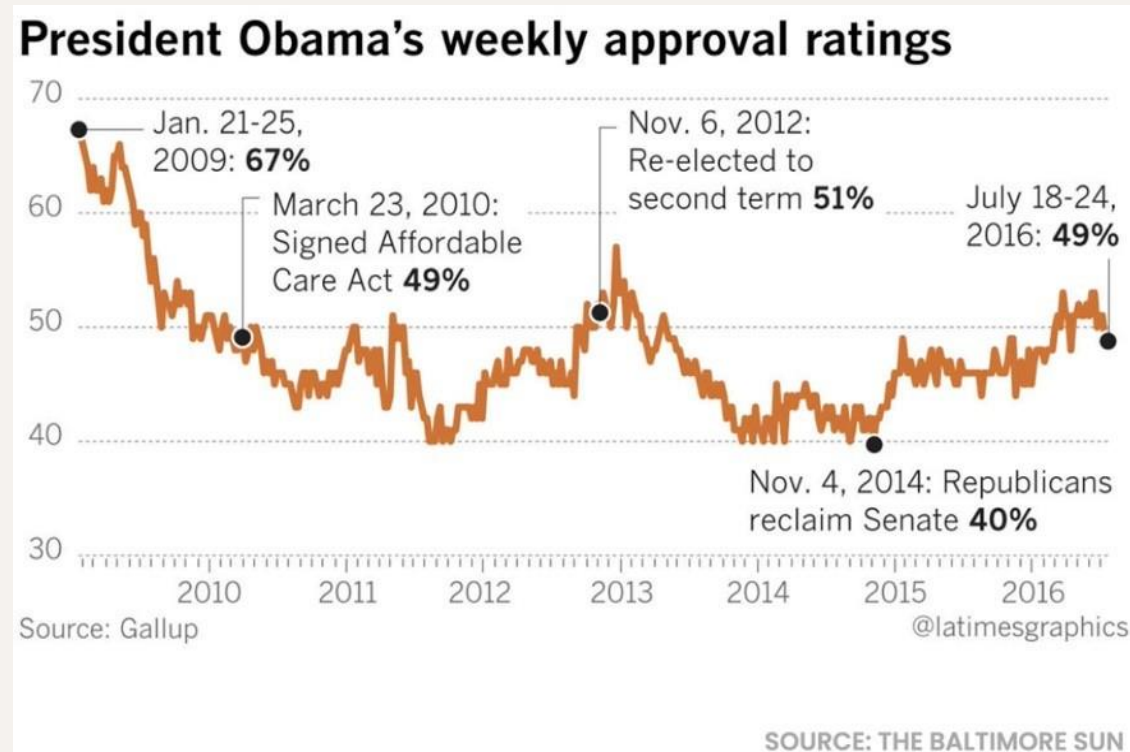
Ethics and data visualization

But what if changing the scale on a bar chart or otherwise manipulating the data better reinforces the point you want to make? Misleading in this manner by inaccurately visualizing data is not OK. Beyond ethical concerns, it is risky territory. All it takes is one discerning audience member to notice the issue (for example, the y -axis of a bar chart beginning at something other than zero) and your entire argument will be thrown out the window, along with your credibility.

Graphical integrity

2 Sometimes we use annotation on the graph to highlight important events.

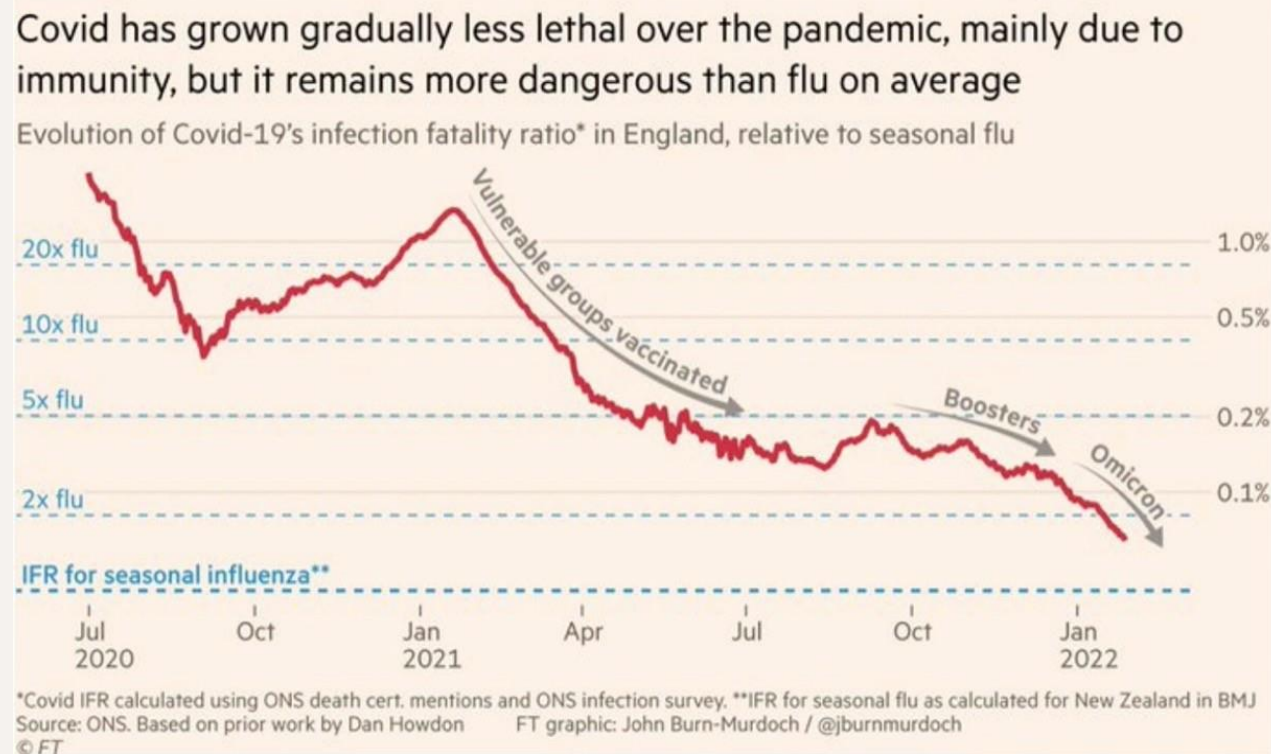
Clear, detailed, and thorough labeling should be used to defeat graphical distortion and ambiguity. Write out explanations of the data on the graphic itself. **Label important events in the data.**



Graphical integrity

2 Sometimes we use annotation on the graph to highlight important events.

Labels provide context and clarity. **Don't label everything.** Only label what has to be labelled.



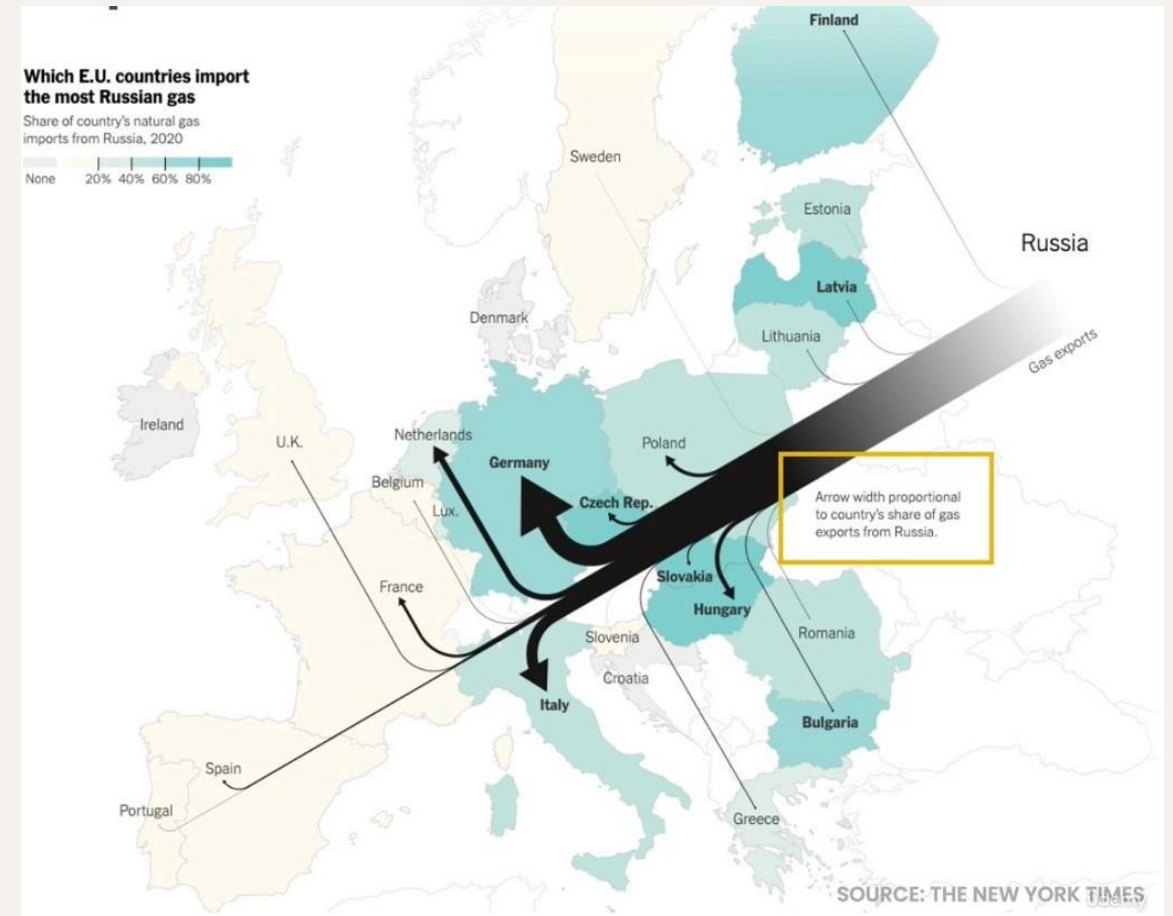
SOURCE: @JBURNMURDOCH

Graphical integrity

2 Sometimes we use annotation on the graph to highlight important events.

Good labelling:

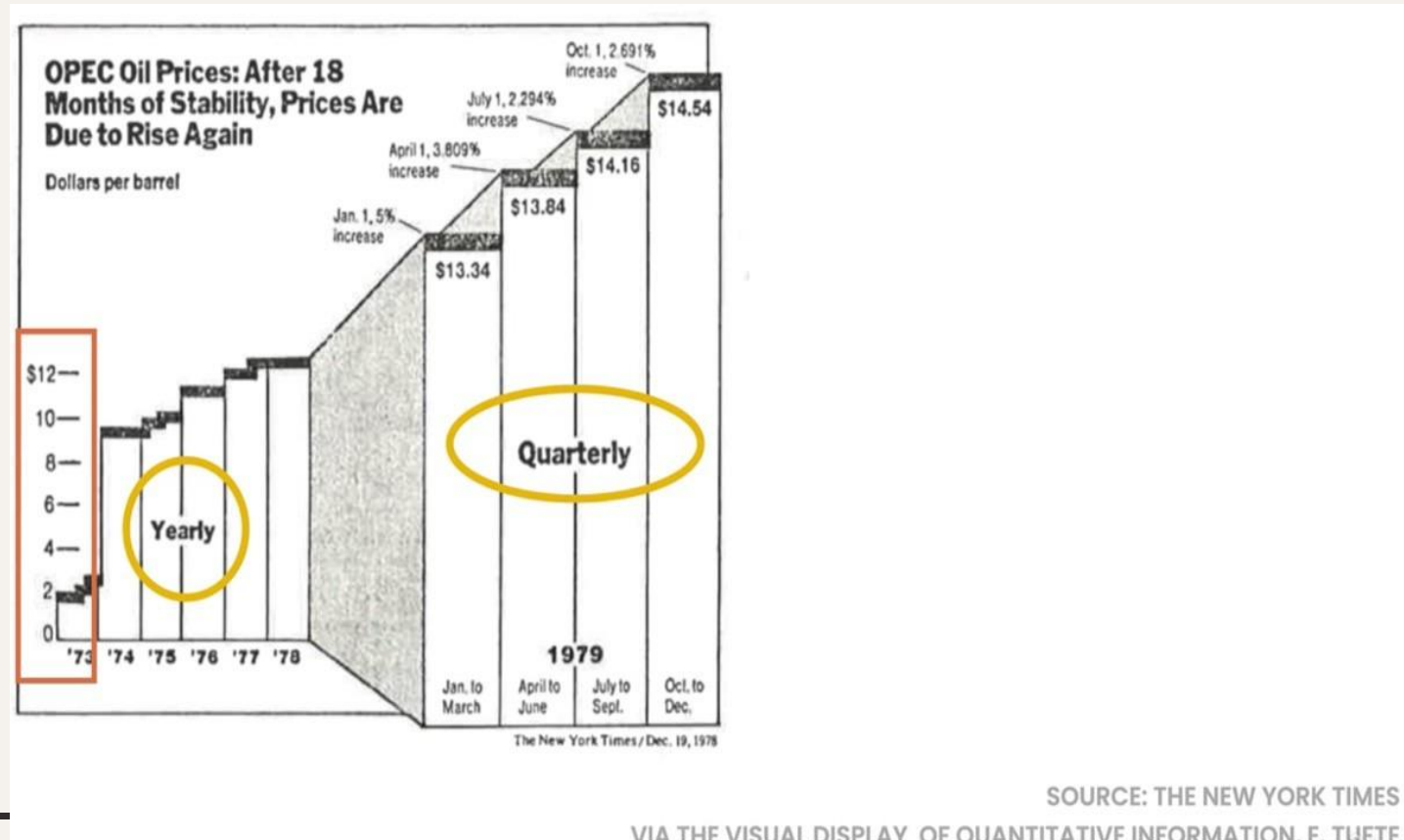
- Explains your data
- Highlight what's important
- Tells your story



Graphical integrity

3 Show Data Variation **Not** Design Variation

While it is not rocket science to make an aesthetic dashboard, sometimes we tend to go overboard with varying designs and colors in a dashboard.

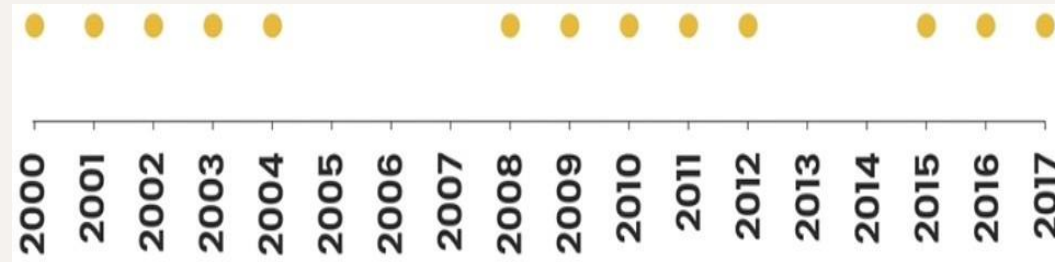




Graphical integrity

4 The number of information showing variables should not exceed the number of dimensions in the data

Years active	
2000	
2001	
2002	
2003	
2004	
2005	
2006	
2010	
2011	
2012	
2015	
2016	
2017	

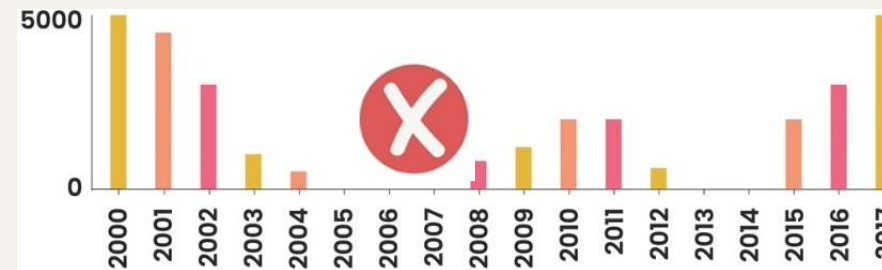
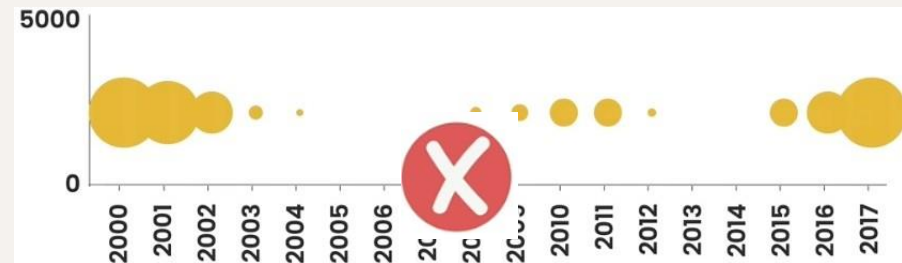




Graphical integrity

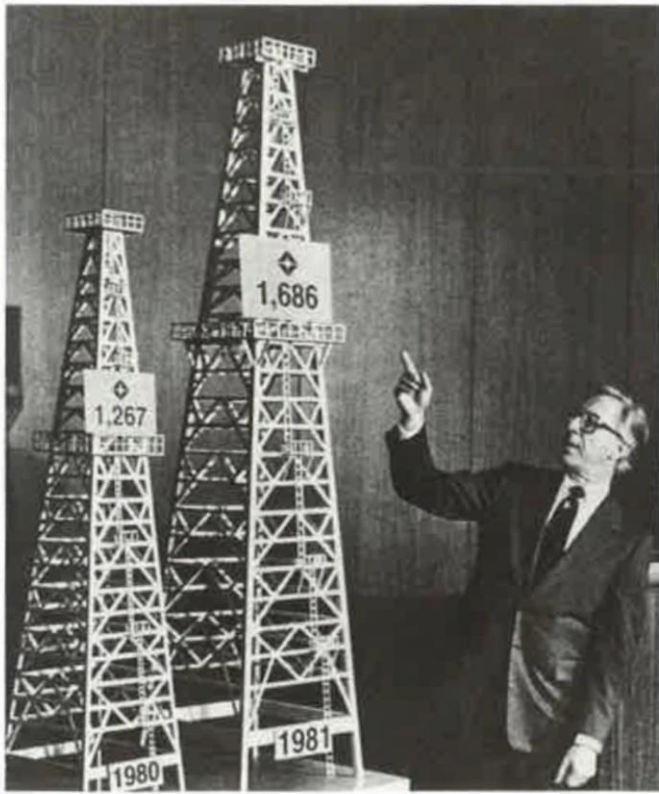
4 The number of information showing variables should not exceed the number of dimensions in the data

Years active	Benefit
2000	5000
2001	4500
2002	3000
2003	1000
2004	500
2005	800
2006	1200
2010	2000
2011	2000
2012	600
2015	2000
2016	3000
2017	5000



Graphical integrity

4 The number of information showing variables should not exceed the number of dimensions in the data



SOURCE: THE NEW YORK TIMES, 1981,

THE VISUAL DISPLAY OF QUANTITATIVE INFORMATION, E. TUFTE

Dimensions of the dataset The number of information showing variables is 2 (Total and Year)

Dimensions of the representation The number of information showing variables is 4 (Total + 3d tower)



In-class activity:

- In groups of 4/5, I will give each group a graph
- Based on the graph, I want you to tell me:
 - 1) What story do you think this chart is trying to tell?
 - 2) Is it telling the story well?
 - 3) How would you improve it? Sketch a corrected version on paper

KNOW YOUR AUDIENCE

A guideline on how personas influence dashboard design



Who is your audience?

A good dashboard is not designed for data, it is designed for a specific user and their decisions.

- **Executives:** Want the headline, the "so what," and the recommendation. Minimal detail. High-level KPIs.
- **Analysts:** Want to dig into the data. Appreciate interactivity, filters, drill-downs. Comfortable with complexity.
- **General public:** Need simple, self-explanatory visuals. No jargon. Clear labels and context.

Different users:

- Have **different goals**
- Make **different decisions**
- Have **different levels of data literacy**



How personas influence dashboard design?

1. Level of detail

Persona	Level of Detail	Example
Executive	High-level overview	Total revenue, growth rate
Manager	Moderate detail	Performance by region
Analyst	High granularity	Transaction-level data



How personas influence dashboard design?

2. Complexity of visuals & interactivity

Persona	Typical Visuals
Executive	KPI cards, simple bar charts
Manager	comparisons, trends
Analyst	scatter plots, decomposition trees

The more analytical the user, the more interaction they need.

Persona	Interaction Level
Executive	Low
Manager	Medium
Analyst	High



How personas influence dashboard design?

3. Time to insight & device

Persona	Expected Time
Executive	5–10 seconds
Manager	30 seconds
Analyst	Several minutes

How quickly the user should understand the message.

Personas also determine **where dashboards are used**.

Device	Design Implications
Desktop	Detailed dashboards
Laptop	Standard BI reports
Mobile	Simplified KPI views



Who is your audience? **Make a decision**

1. In your final groups

2. Draft a 1-paragraph persona for a hypothetical stakeholder for your dataset: Who are they? What's their role? What do they care about? What decision will your data help them make?